Student Community Graphs

Overview and Motivation

Provide an overview of the project goals and the motivation for it. Consider that this will be read by people who did not see your project proposal.

General overview: This project is a survey visualization project, where we collect student data and organize it in a way that we can evaluate clusters and how different clusters inform the opinions of others viewing the charts.

Motivation: WPI Students are extremely invested in various clubs, departments, and Greek life organizations. Many students are looking for resources to find new things that they might enjoy on campus. That is why we create a web of connections between club members on campus to allow for easy recognition of where similar clusters of students tend to congregate. Students would be able to see these graphs and determine what other organizations are popular based on their current involvements and interests. Additionally, we look to determine the best graph visualization methods for different applications, including force-directed graphs, circular link graphs, hierarchical/tree graphs, and bipartite graphs.

Related Work

Anything that inspired you, such as a paper, a web site, visualizations we discussed in class, etc.

For each graph visualization, we used existing code as a starting point. These visualizations from d3 and observablehq.com were helpful in making our graphs visually appealing and readable.

- https://d3js.org/d3-hierarchy/treemap
- https://observablehq.com/@ilyabo/weighted-bipartite-graph
- https://observablehq.com/@d3/force-directed-graph-component
- https://observablehq.com/@d3/chord-diagram

Our inspiration for the project began with the idea of discovering how many people we might be connected to on campus. For many different reasons, collecting this type of data would probably be inappropriate as it would involve lists of names of people we know.

We took this idea of connections and made it more related to groups of individuals as we could accurately create graphs to show interactions between different groups.

Initially, we planned to transform these graphs into a visualization to better understand how you might be able to get more connected with similar groups. However, we settled on representing the same dataset using a variety of methods to evaluate which were most effective in relaying the same information.

Questions

What questions are you trying to answer? How did these questions evolve over the course of the project? What new questions did you consider in the course of your analysis?

Questions we are trying to answer:

1. Given that a WPI student is in x and y club and has z major, what other organizations may they be interested in joining?
2. Which type of visualization is easiest to understand out of force directed graph, circular link graph, bipartite graph, and tree map?
3. How do students value representation of information across the same dataset?

How these questions evolved:

1. This question has stayed relatively the same from the start, since it has been the main question we have set out to answer.
2. This question developed towards the later stages of the project. To start, we were not entirely sure which visualizations we would create. We had always planned on creating force-directed graphs, but the others were discovered through related research.
3. At the start, we tried to answer this question using one graph. Later, we realized more graphs would be more effective, so we evolved to try and answer this question using multiple graphs instead.

New questions considered: The second question was new and something that developed throughout the course of the project and somewhat through trial and error.

Data

Source, scraping method, cleanup, etc.

Data: To source our data, we created a JotForm survey for participants to fill out. This survey was sent out to all WPI seniors, and we got about 130 responses. To download and scape the data, JotForm provides an automatic way to download the survey results in csv format. For cleanup, we had to delete some rows that were not real survey responses, due to our testing at the beginning or due to participants submitting many blank responses. The other issue we dealt with was missing data. Since questions were not mandatory, there was some missing data. However, this was low in quantity, and we were able to continue by leaving the missing data without removing the rows, since we did not have a large number of total responses. Data transformation was simple due to the standardized format produced by JotForm. The only complexity in going from CSV to JSON was the multi-valued fields, which were newline-separated lists.

Additionally, after graph completion we were able to export the Qualtrics interview data to a drive spreadsheet for each question and remove redundant or unnecessary survey information. Specific coding of the data questions was done by inputting the graphics information to more clearly delineate which graph each comment was about. This data is available at the following link:
https://docs.google.com/spreadsheets/d/1A5uPvd5sCCNbq3NmrR9tjlJmMyBybMj-4-H82JOFS2Q/edit?usp=sharing

Exploratory Data Analysis

What visualizations did you use to initially look at your data? What insights did you gain? How did these insights inform your design?

Initial visualizations: The original chart that we explored was the force-directed graph. From there, we discovered similar ways to display the same data, finding and creating a circular link graph, and a bipartite graph. From these visualizations, we were able to answer the questions that are presented above. This informed our design by reevaluating our visualizations to determine if there were any ways to make the graphs easier to understand, especially for someone who does not have a strong background or experience with data visualizations.

Design Evolution

What are the different visualizations you considered? Justify the design decisions you made using the perceptual and design principles you learned in the course. Did you deviate from your proposal?

The visualizations we considered and created are force-directed, circular link, bipartite, and tree map. Using tactics that we learned in the course such as color, node thickness, and organization of charts, we made deliberate decisions to make these complex graphs as easy to understand as possible. Throughout the project, we stayed on course with the vision from our proposal.

Implementation

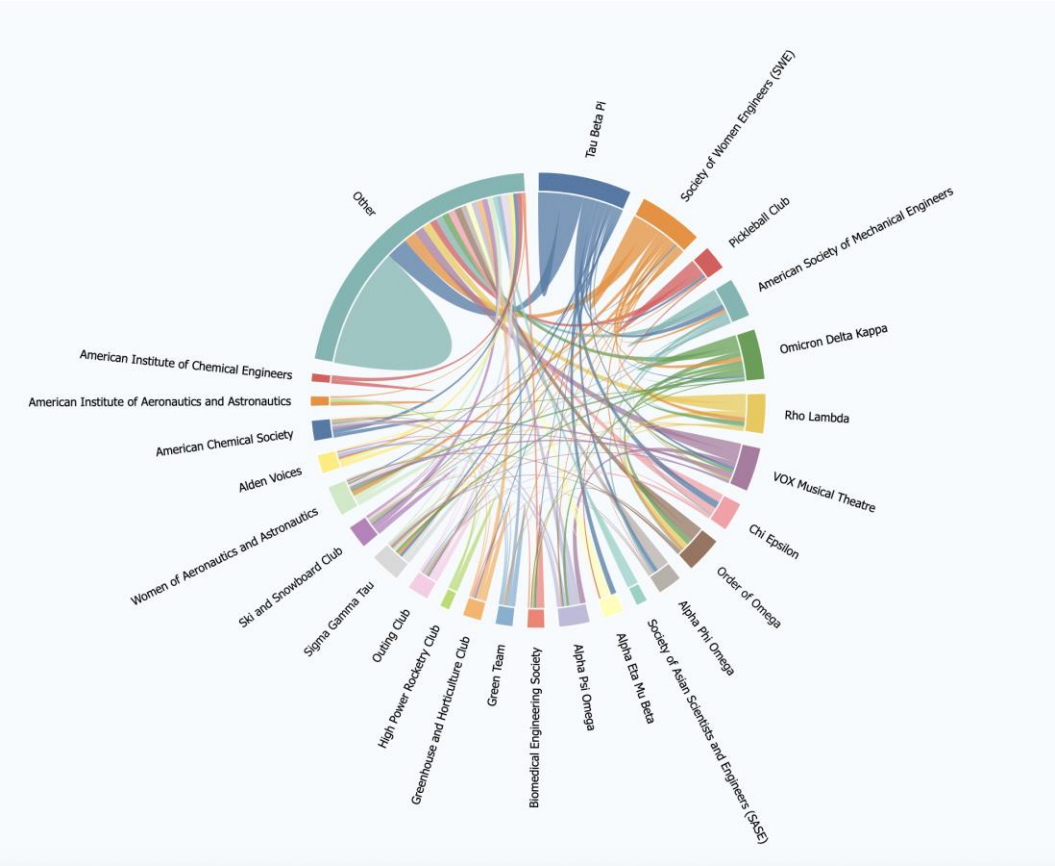
Describe the intent and functionality of the interactive visualizations you implemented. Provide clear and well-referenced images showing the key design and interaction elements.

Force-Directed

The force-directed graph is a series of nodes and edges where the nodes represent different groups, and the edges represent connections between groups, such as members who share both groups. The Force-directed graph is physics simulated, and each point can be dragged around to change the shape of the representation. The example image below is of program connections, in this case the edges represent multiple majors per student.

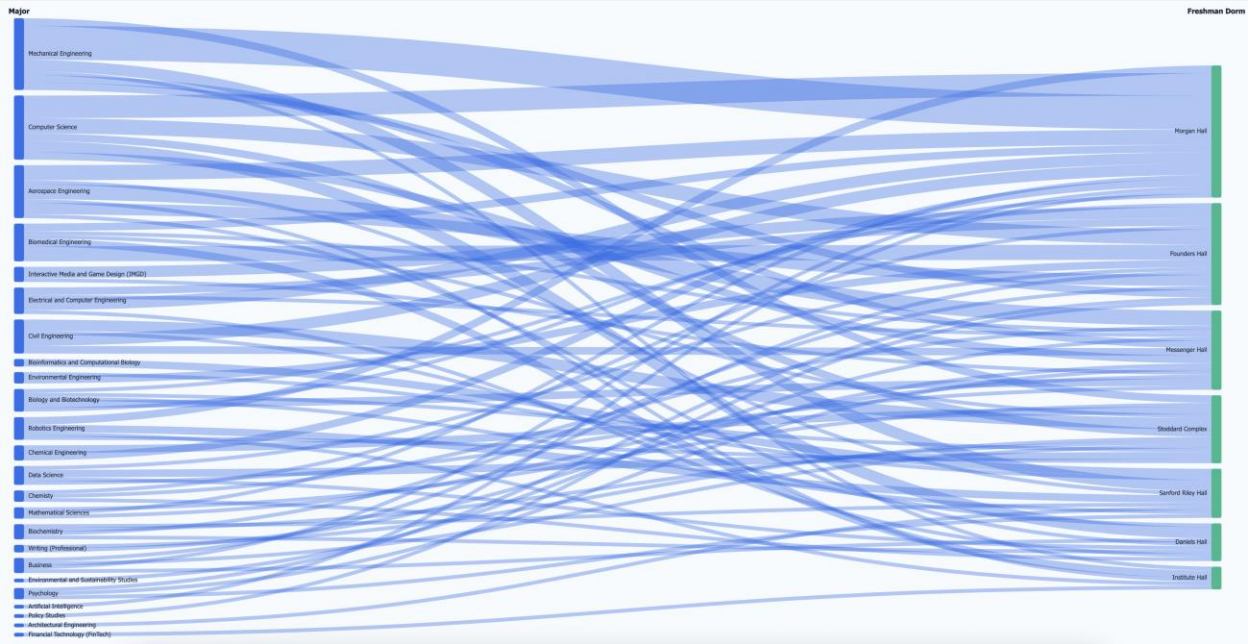
Circular Link

For Circular Link each organization is a node at the edge of a hollow pie chart and each connection is a curved line between the groups. Colors of edges depend on both nodes. Nodes are ordered by side, meaning Tau Beta Pi is the largest with SWE being second largest and AICE being tied for smallest. The Other was an input field. This specific example of the circular link shows organizations and clubs.



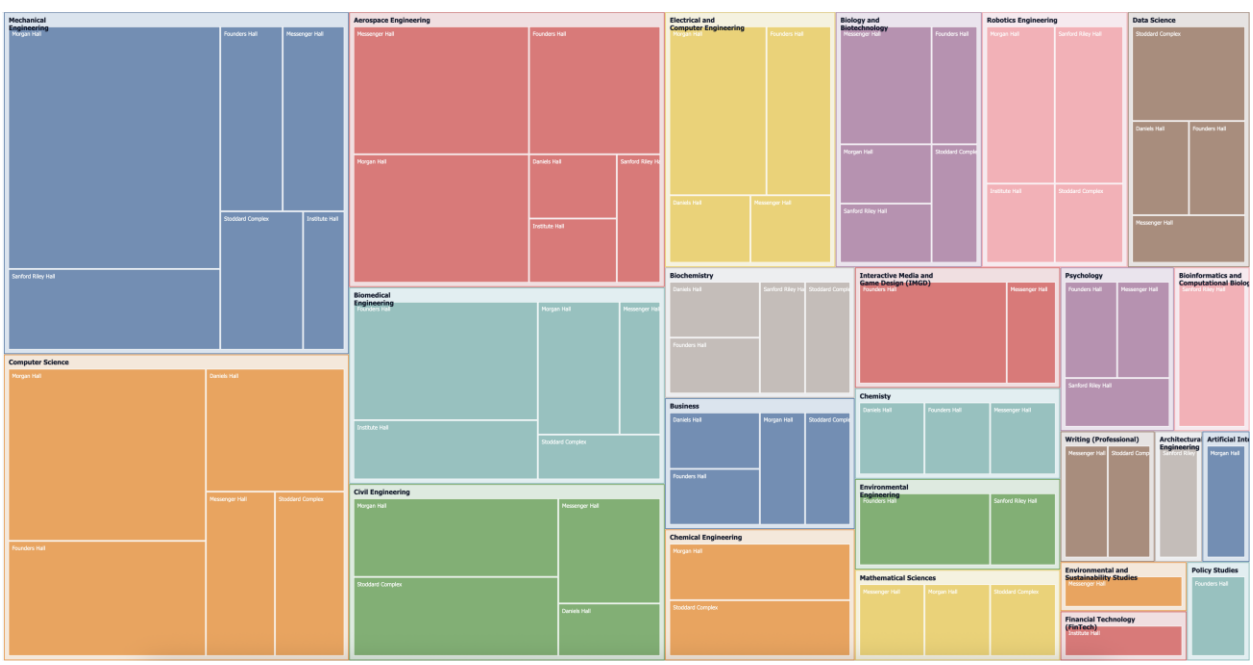
Bipartite

This graph shows interactions between two different classes. This graph does not represent individual connections within a class group but does represent connections across two separate classes. The size of each node in each class is representative of the amount of individuals in that node. In this specific example it is a breakup of majora and freshman dorm assignment.



Tree Map

Finally, the Tree Map is another representation of two different types of classes interacting. Instead of nodes physically connecting to edges, however, the Tree Map makes a representation of comparable sizes of groups. Each class is split into different groups by another class. In this Tree Map example below each Major is distributed by freshman dorm. The specific sizes of both classes in the representation is equal to the proportion of students in that group.



Evaluation

What did you learn about the data by using your visualizations? How did you answer your questions? How well does your visualization work, and how could you further improve it?

Importantly, we learned visualizations of campus involvement require a highly significant amount of data. Despite the fact we got 130 responses to our survey, there was still a high proportion of groups with 1 or 0 representation. This is probably because there are so many programs and clubs on campus that it is hard to track all of them. Curiously, there was also a distinct sampling bias on the amount of representation of different clubs and programs. The makeup of the size of clubs and programs on campus does not necessarily map our graphs closely. These representations might more accurately represent individuals who respond to surveys within each group.

We were able to successfully answer our questions about surveys and student group interactions through our variety of effective visualizations on campus commitments. We additionally were able to receive feedback about the efficacy of our visualizations using an independent survey of people who did not fill out the form information. These responses suggest that the Tree Map was the easiest to understand with the Force-Directed Graph being hardest to understand. In the future, the improvements we could apply to our visualizations would mainly be visual changes of connections and nodes.

Circular Link	
Q31	Q33
How effective was the second graph at representing this data set?	How much did you like the second graph?
{"ImportId":"QID35"}	{"ImportId":"QID37"}
Extremely Effective	Like a great deal
Somewhat Effective	Neither like nor dislike
Somewhat Effective	Like somewhat
Extremely Effective	Like a great deal
Somewhat Effective	Like somewhat
Somewhat Effective	Like somewhat

Treemap	
Q40	Q41
How effective was the second graph at representing this data set?	How much did you like the second graph?
{"ImportId":"QID40"}	{"ImportId":"QID41"}
Extremely Effective	Like a great deal
Somewhat Effective	Dislike somewhat
Somewhat Effective	Like somewhat
Somewhat Effective	Like somewhat
Extremely Effective	Like a great deal
Somewhat Effective	Like somewhat